

JUSTIN CHIH-YAO CHEN

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EDUCATION



The University of North Carolina at Chapel Hill

Ph.D. student in Computer Science

Chapel Hill, NC

Aug. 2023 – Present



National Taiwan University

Master of Data Science. GPA: 4.10 / 4.30

Taipei, Taiwan

Sep. 2020 – Jun. 2022



National Cheng Kung University

Bachelor of Statistics, Double Major in Computer Science. GPA: 3.93 / 4.30

Tainan, Taiwan

Sep. 2016 – Jun. 2020

EXPERIENCE



MURGe-Lab NLP Group

Research Assistant (Advisor: [Mohit Bansal](#))

Improving LLM exploration, efficiency, and multi-agent collaboration for reasoning

Chapel Hill, NC

Aug. 2023 – Present



Microsoft Research

Research Intern (Mentors: [Harsh Jhamtani](#), [Gaurav Verma](#), [Benjamin Van Durme](#))

Learning self-evolving agent skills from user log data

Redmond, WA

May 2026 – Aug. 2026



Salesforce AI Research

Research Intern (Mentors: [Xiangyu Peng](#), [Prafulla Kumar Choubey](#), [Jason Wu](#))

Pushing the boundaries of LLM reasoning via post-training with self-generated hints

Palo Alto, CA

May 2025 – Aug. 2025



Google Cloud AI Research

Student Researcher (Mentors: [Chen-Yu Lee](#), [Zifeng Wang](#))

Enhancing the reasoning of LLMs by strengthening their reverse-thinking capabilities

Sunnyvale, CA

Jun. 2024 – Sep. 2024



Institute of Information Science, Academia Sinica

Research Assistant (Advisor: [Lun-Wei Ku](#))

Working on attractive headline generation, representation learning, and vision question generation

Taipei, Taiwan

Jul. 2020 – Dec. 2022



Institute of Data Science, National Cheng Kung University

Research Assistant (Advisor: [Cheng-Te Li](#))

Working on zero-shot relation extraction and anomaly detection

Tainan, Taiwan

Jun. 2018 – Jun. 2020

PUBLICATIONS (*CO-FIRST AUTHOR)

CITATIONS: 1024, H-INDEX: 13

- [1] **Justin Chih-Yao Chen***, Sukwon Yun*, Elias Stengel-Eskin*, Tianlong Chen, Mohit Bansal. “Skill-Based Mixture-of-Experts: Adaptive Routing for Heterogeneous Reasoning via Inferred Skills.” **ICML 2026**. [\[Paper\]](#) [\[Code\]](#)
- [2] **Justin Chih-Yao Chen**, Becky Xiangyu Peng, Prafulla Kumar Choubey, Kung-Hsiang Huang, Jiaxin Zhang, Mohit Bansal, Chien-Sheng Wu. “Nudging the Boundaries of LLM Reasoning.” **ICLR 2026**. [\[Paper\]](#) [\[Code\]](#)
- [3] **Justin Chih-Yao Chen**, Zifeng Wang, Hamid Palangi, Rujun Han, Sayna Ebrahimi, Long Le, Vincent Perot, Swaroop Mishra, Mohit Bansal, Chen-Yu Lee, Tomas Pfister. “Reverse Thinking Makes LLMs Stronger Reasoners.” **NAACL 2025**. [\[Paper\]](#) [\[Code\]](#)
- [4] **Justin Chih-Yao Chen**, Archiki Prasad, Swarnadeep Saha, Elias Stengel-Eskin, Mohit Bansal. “MAgICoRe: Multi-Agent, Iterative, Coarse-to-Fine Refinement for Reasoning.” **EMNLP 2025**. [\[Paper\]](#) [\[Code\]](#)
- [5] **Justin Chih-Yao Chen**, Swarnadeep Saha, Mohit Bansal. “ReConcile: Round-Table Conference Improves Reasoning via Consensus among Diverse LLMs.” **ACL 2024**. [\[Paper\]](#) [\[Code\]](#)
- [6] **Justin Chih-Yao Chen***, Swarnadeep Saha*, Elias Stengel-Eskin, Mohit Bansal. “MAGDI: Structured Distillation of Multi-Agent Interaction Graphs Improves Reasoning in Smaller Language Models.” **ICML 2024**. [\[Paper\]](#) [\[Code\]](#)

- [7] **Justin Chih-Yao Chen**, Tun-Min Hung, Lun-Wei Ku. “Label-Aware Hyperbolic Embeddings for Fine-grained Emotion Classification.” **ACL 2023**. [\[Paper\]](#) [\[Code\]](#)
- [8] **Justin Chih-Yao Chen**, Dennis Wu, Lun-Wei Ku. “HonestBait: Forward References for Attractive but Faithful Headline Generation.” **ACL Findings 2023**. [\[Paper\]](#)
- [9] **Justin Chih-Yao Chen** and Cheng-Te Li. “ZS-BERT: Towards Zero-Shot Relation Extraction with Attribute Representation Learning.” **NAACL 2021**. [\[Paper\]](#) [\[Code\]](#)
- [10] **Justin Chih-Yao Chen**, Archiki Prasad, Zaid Khan, Joykirat Singh, Runchu Tian, Elias Stengel-Eskin, Mohit Bansal. “Cog-DRIFT: Exploration on Adaptively Reformulated Instances Enables Learning from Hard Reasoning Problems.” arXiv preprint. [\[Paper\]](#) [\[Code\]](#)
- [11] Hyunji Lee*, **Justin Chih-Yao Chen***, Joykirat Singh, Zaid Khan, Elias Stengel-Eskin, Mohit Bansal. “MINTEval: Evaluating Memory under Multi-Target Interference in Long-Horizon Agent Systems.” arXiv preprint. [\[Paper\]](#) [\[Code\]](#)
- [12] Jaewoo Lee, Archiki Prasad, **Justin Chih-Yao Chen**, Zaid Khan, Elias Stengel-Eskin, Mohit Bansal. “PRInTS: Reward Modeling for Long-Horizon Information Seeking.” **ACL 2026**.
- [13] Tianyi Niu, **Justin Chih-Yao Chen**, Genta Indra Winata, Shi-Xiong Zhang, Supriyo Chakraborty, Sambit Sahu, Yue Zhang, Elias Stengel-Eskin, Mohit Bansal. “Routing with Generated Data: Annotation-Free LLM Skill Estimation and Expert Selection.” **ACL 2026**.
- [14] Joykirat Singh, **Justin Chih-Yao Chen**, Archiki Prasad, Elias Stengel-Eskin, Akshay Nambi, Mohit Bansal. “Think Right: Learning to Mitigate Under-Over Thinking via Adaptive, Attentive Compression.” **COLM 2026**.
- [15] Nithin Sivakumaran, **Justin Chih-Yao Chen**, David Wan, Yue Zhang, Jaehong Yoon, Elias Stengel-Eskin, Mohit Bansal. “DART: Leveraging Multi-Agent Disagreement for Tool Recruitment in Multimodal Reasoning.” **EACL 2026**.
- [16] Cheng-Te Li, Yu-Che Tsai, **Justin Chih-Yao Chen**, Jay Chiehen Liao. “Graph Neural Networks for Tabular Data Learning: A Survey with Taxonomy and Directions.” **ACM Computing Surveys 2025**.
- [17] David Wan, **Justin Chih-Yao Chen**, Elias Stengel-Eskin, Mohit Bansal. “MAMM-Refine: A Recipe for Improving Faithfulness in Generation with Multi-Agent Collaboration.” **NAACL 2025**.
- [18] Swarnadeep Saha, Archiki Prasad, **Justin Chih-Yao Chen**, Peter Hase, Elias Stengel-Eskin, Mohit Bansal. “System-1.x: Learning to Balance Fast and Slow Planning with Language Models.” **ICLR 2025**.
- [19] Nicholas Suwono, **Justin Chih-Yao Chen**, Tun-Min Hung, Ting-Hao Huang, I-Bin Liao, Yung-Hui Li, Lun-Wei Ku, Shao-Hua Sun. “Location-Aware Visual Question Generation with Lightweight Models.” **EMNLP 2023**.

REFERENCES

Mohit Bansal Director of the MURGe-Lab, UNC Chapel Hill mbansal@cs.unc.edu	<i>PhD advisor</i>	Jason Wu Senior Director, Salesforce AI Research wu.jason@salesforce.com	<i>Internship mentor</i>
Chen-Yu Lee Principal Scientist and Director, Google Cloud AI Research chenyulee@google.com	<i>Internship mentor</i>	Hamid Palangi Staff Research Scientist, Google; Associate Professor, UW hamidpalangi@google.com	<i>Internship mentor</i>
Zifeng Wang Research Scientist, Google Cloud AI Research zifengwangking@gmail.com	<i>Internship mentor</i>	Harsh Jhamtani Principal Research Scientist, Copilot Tuning, Microsoft hjhamtani@microsoft.com	<i>Internship mentor</i>
Gaurav Verma Applied Scientist, IDEAS Research, Microsoft gauravverma@microsoft.com	<i>Internship mentor</i>	Benjamin Van Durme Professor, JHU; Research Manager, Microsoft vandurme@jhu.edu	<i>Internship mentor</i>
Lun-Wei Ku Director of the NLP/SA Lab, Institute of Information Science, Academia Sinica lwku@iis.sinica.edu.tw	<i>Master’s advisor</i>	Cheng-Te Li Director of the Networked AI Lab, National Cheng Kung University chengte@ncku.edu.tw	<i>Undergraduate advisor</i>